Unitary Representations of the Galilei Group

In non-relativistic physics, inertial frames are related by Galilean transformations. If two observers are in inertial frames, their spacetime coordinates are related by

$$(\mathbf{x}, t) \mapsto (R\mathbf{x} + \mathbf{u}t + \mathbf{a}, t + s)$$

These transformations form a continuous group: the Galilei group. Since the laws of non-relativistic physics are invariant under these transformations, any quantum theory describing such physics must realize this symmetry. This is the core principle underlying our correspondence rules, whereby we identify particular dynamical variables with particular operators.

By Wigner's theorem, any continuous symmetry transformation connected to the identity must be induced by a unitary operator acting on the underlying Hilbert space. Galilean transformations are continuous and connected to the identity, so they are represented by unitary operators. Thus, for each Galilean transformation g there is a unitary operator $U(g)$. Every continuous one-parameter subgroup gives a unitary family

$$U(\epsilon) = \exp\left(-\frac{i}{\hbar}\epsilon K\right), \tag{1}$$

where K is a self-adjoint operator called the generator of the transformation, and ϵ is a continuous parameter. For the Galilei group, the generators are:

Transformation	**Unitary Operator**	**Generator**
Spatial translation ($\mathbf{a}$)	$U(\mathbf{a}) = \exp\left(-\frac{i}{\hbar}\,\mathbf{a}\cdot\mathbf{p}\right)$	$\hat{p}_i$ (momentum)
Time translation (s)	$U(s) = \exp\left(\frac{i}{\hbar}\,sH\right)$	$-H$ ($-$Hamiltonian*)
Rotation ($R(\boldsymbol{\theta})$)	$U(R) = \exp\left(-\frac{i}{\hbar}\,\boldsymbol{\theta}\cdot\mathbf{J}\right)$	J_i (angular momentum)
Galilean boost ($\mathbf{u}$)	$U(\mathbf{u}) = \exp\left(-\frac{i}{\hbar}\,\mathbf{u}\cdot\mathbf{G}\right)$	G_i (boost generator)

Our goal is to capture the group structure in the operator representations. Any sequence of space–time transformations is equivalent to another transformation in the group, so the composition of unitary transformations must respect this property. In fact, since physical states are only defined up to an overall phase, the product of two unitary operators implementing group elements may differ from the operator for the composed group element only by a phase. Generally speaking, this means

$$U(g_1)U(g_2) = e^{i\gamma(g_1,g_2)}U(g_1 \circ g_2) \tag{2}$$

for some real-valued function $\gamma(g_1, g_2)$. In many cases, we can determine the phase by additional considerations. Curiously, we cannot simply choose $\gamma = 0$ in all cases.

For example, consider a translation followed by a boost. Since the transformations commute with one another, at the level of their unitary representations, we may write, e.g.,

$$e^{-\frac{i}{\hbar}uG_j}e^{-\frac{i}{\hbar}a\hat{p}_i} = e^{i\gamma(a,u)}e^{-\frac{i}{\hbar}a\hat{p}_i}e^{-\frac{i}{\hbar}uG_j}$$

and if we multiply from the right by the inverse boost followed by the inverse translation,

$$e^{-\frac{i}{\hbar}uG_j}e^{-\frac{i}{\hbar}a\hat{p}_i}e^{\frac{i}{\hbar}uG_j}e^{\frac{i}{\hbar}a\hat{p}_i} = e^{i\gamma(a,u)}$$

For infinitesimal a and u, we expand the left-hand-side to leading non-trivial order (tedious):

$$e^{-\frac{i}{\hbar}uG_j}e^{-\frac{i}{\hbar}a\hat{p}_i}e^{\frac{i}{\hbar}uG_j}e^{\frac{i}{\hbar}a\hat{p}_i} \approx \mathbb{1} + \frac{au}{\hbar^2}[\hat{p}_i, G_j]$$

Hence, to leading order in a and u, we find

$$e^{i\gamma} = \exp\left(\frac{au}{\hbar^2}[\hat{p}_i, G_j]\right),$$

which holds if and only if $[\hat{p}_i, G_j] = ic_{ij}\mathbb{1}$ for real-valued constants c_{ij}. In other words, although the transformations commute with one another, the corresponding unitary operators commute only up to an overall phase, and the most general conclusion we can make is that the commutator of the generators is proportional to the identity operator.

A Spinless Particle in 3D Space

Up to this point, the generators have been introduced and constrained purely from their role in implementing symmetry transformations. This approach is powerful but abstract. To make contact with measurable quantities, it is natural to shift perspective. Instead of starting from transformations, we now begin with what can actually be observed. A fundamental example is the *position* of a particle. Experimentally, a particle's position can be measured, and each possible outcome corresponds to a definite point in space. In accordance with Lecture #8, we therefore introduce position eigenstates $|\mathbf{r}\rangle$ labeled by continuous eigenvalues, with the understanding that the particle carries no internal degrees of freedom (e.g., spin).

Evaluation of Commutators

In Lecture #4, we established the rotation commutation relations:

$$[J_i, A_j] = i\hbar\epsilon_{ijk}A_k \tag{3}$$

for any operator-valued 3-vector (or vector operator) $\mathbf{A}$, e.g., $\mathbf{r}, \mathbf{p}, \mathbf{G}$, and $\mathbf{J}$ itself. In Lecture #8, we established the following canonical commutation relations, along with a few others:

$$[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij}, \quad \text{and} \quad [\hat{r}_i, \hat{r}_j] = [\hat{p}_i, \hat{p}_j] = 0. \tag{4}$$

The commutation relations above will help us determine the constants c_{ij} in the relation $[\hat{p}_i, G_j] = ic_{ij}\mathbb{1}$ we found earlier. From the Jacobi identity for the triplet $(J_1, G_2, \hat{p}_1)$,

$$[J_1, [G_2, \hat{p}_1]] + [G_2, [\hat{p}_1, J_1]] + [\hat{p}_1, [J_1, G_2]] = 0,$$

$$[\hat{p}_1, G_3] = 0, \text{ and likewise, } [\hat{p}_i, G_j] = 0 \text{ for } i \neq j$$

From the Jacobi identity for $(J_1, G_2, \hat{p}_3)$,

$$[J_1, [G_2, \hat{p}_3]] + [G_2, [\hat{p}_3, J_1]] + [\hat{p}_3, [J_1, G_2]] = 0,$$

$$i\hbar[G_2, \hat{p}_2] + i\hbar[\hat{p}_3, G_3] = 0 \Rightarrow [G_2, \hat{p}_2] = [G_3, \hat{p}_3],$$

and by extension, $[G_1, \hat{p}_1] = [G_2, \hat{p}_2] = [G_3, \hat{p}_3]$. Combining results from above, we find

$$[\hat{p}_i, G_j] = i\hbar\delta_{ij}m\mathbb{1} \tag{5}$$

where m is a real constant and it turns out $m = 0$ if and only if $\mathbf{G} = 0$.

The commutation relations severely restrict the possible operator structure. The following theorems will help us establish key relations between our symmetry generators, which will enable us to connect them with physical observables.

Theorem 1. Any operator that commutes with $(\hat{r}_1, \hat{r}_2, \hat{r}_3)$ must be a function of the operators in that set, and likewise for $(\hat{p}_1, \hat{p}_2, \hat{p}_3)$.

Theorem 2. For a particle with no internal degrees of freedom, any operator that commutes with $\mathbf{r}$ and $\mathbf{p}$ must be a multiple of the identity[1].

[1] See Ballentine, 1998.

Physical Observables

From Eq. (5) and the canonical commutation relations, it follows that

$$[\hat{p}_i, G_j] = mi\hbar\delta_{ij} = m[\hat{r}_j, \hat{p}_i] = [\hat{p}_i, -m\hat{r}_j]$$

$$[\hat{p}_i, G_j + m\hat{r}_j] = 0$$

For the instantaneous boost transformation at $t = 0$, the position is unaffected. Hence, $[\hat{r}_i, G_j] = 0$, and furthermore,

$$[\hat{r}_i, G_j + m\hat{r}_j] = 0$$

Each component of $\mathbf{G} + m\mathbf{r}$ commutes with the components of $\mathbf{r}$ and $\mathbf{p}$, so by Theorem 2, we conclude $\mathbf{G} = -m\mathbf{r} + \mathbf{c}\mathbb{1}$ where $\mathbf{c}$ is a constant vector. However, since $\mathbf{G}$ is a vector operator, it must satisfy the rotation commutation relations, which forces $\mathbf{c} = 0$. Hence,

$$\mathbf{G} = -m\mathbf{r} \quad (t = 0) \tag{6}$$

From the Heisenberg equation of motion, we find

$$\frac{d\langle\mathbf{r}\rangle}{dt} = \frac{i}{\hbar}\Big\langle [H, \mathbf{r}] \Big\rangle$$

and in light of this result, we define the *velocity operator* as

$$\mathbf{v} = \frac{i}{\hbar}[H, \mathbf{r}] \tag{7}$$

The velocity operator must transform under a unitary boost operation in a way that represents the transformation $\mathbf{v} \to \mathbf{v} + \mathbf{u}$. Hence,

$$e^{\frac{i}{\hbar}\mathbf{u}\cdot\mathbf{G}}\mathbf{v}e^{-\frac{i}{\hbar}\mathbf{u}\cdot\mathbf{G}} = \mathbf{v} + \mathbf{u}, \tag{8}$$

Expanding the unitary transformation to leading order in u_i leads to

$$[\hat{v}_i, G_j] = i\hbar\delta_{ij} \tag{9}$$

which is analogous to the canonical commutation relations if you consider a boost to be a *translation in velocity space*. From Eq. (5), it follows that

$$[\hat{v}_i, G_j] = i\hbar\delta_{ij} = [\hat{p}_i/m, G_j]$$

$$[\hat{v}_i - \hat{p}_i/m, G_j] = 0 \Rightarrow [\hat{v}_i - \hat{p}_i/m, \hat{r}_j] = 0$$

Since the relation above holds for all j, the quantity $\hat{v}_i - \hat{p}_i/m$ must be a function of $\mathbf{r}$ by Theorem 1. Hence,

$$\hat{v}_i - \frac{\hat{p}_i}{m} = f(\mathbf{r}) = -\frac{A_i(\mathbf{r})}{m} \tag{10}$$

or

$$\mathbf{v} = \frac{\mathbf{p} - \mathbf{A}(\mathbf{r})}{m} \tag{11}$$

where $\mathbf{A}(\mathbf{r})$ is called a *vector potential*, and it's typically relevant when interactions are velocity-dependent yet energy-conserving.

From the definition of the velocity operator,

$$\mathbf{v} = \frac{i}{\hbar}[H, \mathbf{r}] = \frac{\mathbf{p} - \mathbf{A}(\mathbf{r})}{m}$$

Recall that $[f, \hat{x}] = -i\hbar \frac{\partial f}{\partial p_x}$ from homework #5. This suggests the commutator with $\mathbf{r}$ acts as a gradient operator over momentum. With this intuition, we simply define the operator

$$H_0 \equiv \frac{\left[\mathbf{p} - \mathbf{A}(\mathbf{x})\right]^2}{2m} = \frac{1}{2m}\sum_j (\hat{p}_j - A_j)(\hat{p}_j - A_j) \tag{12}$$

and now we compute its commutator with $\hat{r}_i$:

$$\begin{aligned}
[H_0, \hat{r}_i] &= \frac{1}{2m}\sum_j \Big[(\hat{p}_j - A_j)[\hat{p}_j - A_j, \hat{r}_i] + [\hat{p}_j - A_j, \hat{r}_i](\hat{p}_j - A_j)\Big] \\
&= \frac{1}{2m}\sum_j \Big[(\hat{p}_j - A_j)[\hat{p}_j, \hat{r}_i] + [\hat{p}_j, \hat{r}_i](\hat{p}_j - A_j)\Big] \\
&= \frac{1}{2m} \cdot -2i(\hat{p}_i - A_i) = -i\left(\frac{\hat{p}_i - A_i}{m}\right) = -i\hat{v}_i \\
&= [H, \hat{r}_i] \\
0 &= [H - H_0, \hat{r}_i]
\end{aligned}$$

Since $H - H_0$ commutes with $\mathbf{r}$, it must be a function of $\mathbf{r}$ by Theorem 1; let's call it $V(\mathbf{r})$. The Hamiltonian can therefore be written as

$$H = H_0 + V(\mathbf{r}) = \frac{\left[\mathbf{p} - \mathbf{A}(\mathbf{r})\right]^2}{2m} + V(\mathbf{r}) \tag{13}$$

where $V(\mathbf{r})$ is a *scalar potential*. Adding a constant multiple of the identity to the Hamiltonian does not affect any commutators with observables, and therefore leaves the dynamics unchanged. Hence, the Hamiltonian is defined only up to an additive constant, which we absorb into the definition of $V(\mathbf{r})$. With this result, we have deduced that the only forms of interaction consistent with invariance under the Galilei group of transformations are a *scalar potential* $V(\mathbf{r})$ and a *vector potential* $\mathbf{A}(\mathbf{r})$. Both of these may be time-dependent. As operators they may be functions of $\mathbf{r}$ but must be independent of $\mathbf{p}$.

Exercises:

1. Recall the commutation relations $[\hat{p}_i, G_j] = i\hbar\delta_{ij} m\mathbb{1}$. For the instantaneous boost transformation at $t = 0$, show that $m = 0$ if and only if $\mathbf{G} = 0$.

2. Derive the following result and remark on its significance:

$$\frac{i}{\hbar}[H, m\hat{v}_j] = -\frac{\partial V}{\partial r_j} - \frac{1}{2}\left[\hat{v}_i\left(\frac{\partial A_j}{\partial r_i} - \frac{\partial A_i}{\partial r_j}\right) + \left(\frac{\partial A_j}{\partial r_i} - \frac{\partial A_i}{\partial r_j}\right)\hat{v}_i\right]$$

   Hint: the right-hand-side is *almost* the same expression as the Lorentz force from an electromagnetic field on a moving charge!

3. Find the operator $\mathbf{G}$ that yields the full Galilei transformation ($t \neq 0$):

$$e^{\frac{i}{\hbar}\mathbf{u}\cdot\mathbf{G}}\mathbf{v}e^{-\frac{i}{\hbar}\mathbf{u}\cdot\mathbf{G}} = \mathbf{v} + \mathbf{u}, \qquad e^{\frac{i}{\hbar}\mathbf{u}\cdot\mathbf{G}}\mathbf{r}e^{-\frac{i}{\hbar}\mathbf{u}\cdot\mathbf{G}} = \mathbf{r} + \mathbf{u}t$$

4. Space is invariant under the scale transformation $\mathbf{r} \to e^c\mathbf{r}$ where c is a real parameter. The corresponding unitary operator may be written as $e^{-icD/\hbar}$, where D is the dilation generator. Determine the commutator $[D, \mathbf{p}]$ between the generators of dilation and space displacements.